Code No: 136FC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
FUNDAMENTALS OF ROBOTICS
(Common to CE, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define manipulator.	[2]
b)	Explain the uses of sensors in robotics.	[3]
c)	List out the classification of grippers.	[2]
d)	What are the limitations of pneumatic actuators?	[3]
e)	What is homogeneous transformation?	[2]
f)	What is the difference between forward and inverse kinematics?	[3]
g)	Draw the block diagram of trajectory planning.	[2]
h)	What are the different types of control modes used in robot system?	[3]
i)	List out the various robot programming methods.	[2]
j)	Describe robot applications in manufacturing.	[3]

PART – B

(50 Marks)

2.a) What are the different ways of classifying the Robots? Explain them in brief.
b) What are the different types of sensors used in robots? Explain the suitable sensor to measure the position of the gripper accurately. [5+5]

OR

3.a) Explain the classification of robots by different controlling methods.
b) Define work volume and load carrying capacity with reference to robot. [5+5]

4.a) Explain the types of gripper mechanisms and force analysis in grippers
b) Explain the applications of robots in detail for various assembly and inspection operations in an industrial environment. [5+5]

OR

5.a) With a neat sketch, explain the magnetic gripper and list its advantages and limitations.
b) What are future manufacturing applications of robot? [5+5]

6.a) Derive the forward kinematics equation using the DH convention for the two-link manipulator with one revolute and one prismatic joint.

b) Using the above formula find the motion parameters for the point a(uvw)= (6,2,4) T rotate 30^0 about the y-axis followed by translation of 6 units along X-axis. [5+5]

OR

7.a) Write the homogeneous transformation matrix for a rotation of 90^0 about the z-axis, followed by a rotation of 90^0 about the x-axis, followed by a translation of (3, 7, 9).

b) Explain the inverse kinematics for any manipulator based on D-H convention. [5+5]

8. Explain tasks involved in trajectory planning. Explain the Cartesian space techniques used for trajectory planning. [10]

OR

9.a) What are the considerations that are made while planning a joint interpolated motion trajectory?

b) With the help of diagram, show the elements of feedback control system. [5+5]

10.a) Discuss the structure, elements and functions of any one Robot Language.

b) Differentiate Manual lead through and powered lead through in robot programming. [5+5]

OR

11.a) Discuss about the WAIT, SIGNAL, DELAY commands.

b) Discuss the relative merits and demerits of different textual robot languages. [5+5]

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